Estimation of a flaw size distribution from a data mixture from destructive tests and non-destructive in-service inspections

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Abstract: Among the stochastic inputs entering into a probabilistic calculation of the reliability of industrial equipments/components of nuclear plants, the flaw size is often a key variable whose distribution has to be carefully estimated. Although the considered Pressurized Water Reactor (PWR) component flaws appear as manufacturing defects and remain non-evolutive with running time, the assessment of this distribution is usually not trivial. Indeed, the observations that may be used to this aim are typically data coming from nondestructive experiments (NDE), affected by observational noise and progressive censoring due to the detection limits of the testing process. The mixture of these data with observations coming from destructive experiments provides a frame to both estimate both the distribution by maximum likelihood techniques and precise the features of the probability of detection (POD).

Keywords: destructive and non-destructive testings, probability of detection, noisy experiments

1. INTRODUCTION

Risk studies on highly reliable industrial components are often limited by the very low number of observed failures. Alternatively, the probability of a failure can be computed through numerical experimentations involving computer codes reproducing the physical (e.g., thermomecanical) behavior of the studied component in its running environment. To explore exhaustively the configurations in the inputs of such numerical codes driving to the occurence of a failure, these inputs are typically considered as stochastic. In many structural reliability studies led on the French nuclear park, a key input is the distribution of flaw sizes. In the peculiar context of PWR steel passive components such flaws appear as manufacturing defects, remain non-evolutive with time and their size can be summarized by a couple of height and width.

Assessing this distribution is usually the solution of a probabilistic inverse problem, associated with indirect noisy measurements from non-destructive in-service inspections. In the present case, the inspections were led by ultrasonic devices. These inspections are characterized by an observational noise and a probability of detection (POD) related to the performance of the examination process, depending mainly of the limiting flaw size (its height). In the problem considered here, a second source of data is available: destructive testing produces data which are statistically considered as direct measures of flaw sizes.

In this article, we establish a general framework for the statistical estimation by maximum likelihood of the flaw size distribution, based on such hypotheses. Numerical integration and asymptotic statistics are used to provide estimators and confidence regions. The methodology can however suffer from the difficulties occuring when calibrating the POD. Checking for the relevance of this POD can be done using the two sources of data. These steps are tested on data characterizing the flaw size of the steel components evoked hereinbefore.

Remark. All flaw size data and value domains used in this article have been rescaled such that they belong to a [0-10] interval.

2. MODELLING

Denote X the random variable associated to a flaw height. The width of a flaw appear significantly larger than its height, so that it can be considered as not affected by observational noise. Therefore we mainly focus on the marginal distribution of X in this paper. Previous studies, led on data coming from destructive experiments, have shown that a relevant modelling for X is the Weibull distribution with probability density function (pdf):

$$f(x|\eta,\beta) \quad = \quad \beta/\eta\,(x/\eta)^{\beta-1}\exp(-(x/\eta)^{\beta})\mathbf{1}_{\{x>0\}}.$$

The parameter vector $\theta=(\eta,\beta)$ must also be estimated from the data. Denote Z the random variable associated to the observation of X through an inspection process, and assume to have noisy observations $\mathbf{z}=(z_1,\ldots,z_n)$ such that $z_i = x_i \cdot e_i$ where, whatever $i=1,\ldots,n$, the multiplicative observational error e_i follows a known distribution with pdf $f_\epsilon(\cdot)$. If the inspection were *destructive* (testing), the observational noise could be considered negligible (ie., $e_i \equiv 1$). For *non-destructive* inspections, e_i follows a Gamma distribution centered on 1 (or, nearly equivalently, a truncated Gaussian distribution), with a fixed coefficient of variation ρ. Information about ρ is typically provided by manufacturer specifications and performance testings of ultrasonic processes. The choice of a multiplicative noise is explained by the positivity of all quantities involved.

Furthermore, it is assumed there exists a unknown number m of missing measures $(z_{n+1},\ldots,z_{n+m})$, *upperly censored* by the detection limits of the measurement process, so that the full vector of latent variables is $\mathbf{x}=(x_1,\ldots,x_{n+m})$. Denote then D the binary random variable saying whether X was detected or not. It depends on the value of X through a function defined in $[0,1]$ as the *probability of detection* (POD):

$$\mathrm{POD}(x) \quad = \quad P(D=1|X=x).$$

In the following, $\mathrm{POD}(x)$ is assumed to be an increasing known function of x. The unconditional POD is given by

$$P_f(\theta) \quad = \quad P(D=1|\theta) \quad = \quad \int_{x\in\mathbb{R}} \mathrm{POD}(x) f(x|\theta)\mathrm{d}x. \tag{1}$$

Besides, from Celeux et *al.* [1],given the number of observation n and θ, m follows a negative binomial distribution $\mathcal{NB}(n,1-P_f(\theta))$:

$$P(M=m|\theta) \quad = \quad \frac{(n+m-1)!}{(n-1)!m!}P_f(\theta)^n(1-P_f(\theta))^m. \tag{2}$$

The vector of parameters θ may then be estimated by maximizing the likelihood of observations. If $m,\mathbf{z}$ and $\mathbf{x}$ were observed, the complete likelihood of the model would be:

$$\begin{aligned}
\ell(m,\mathbf{z},\mathbf{x}|\theta) &= P(M=m|\theta)p(\mathbf{x}|\theta,m)p(\mathbf{z}|\mathbf{x}) \\
&= \frac{(m-1)!}{(n-1)!(m-n)!}P_f(\theta)^n(1-P_f(\theta))^{m-n} \\
&\times \prod_{i=1}^{n}\frac{\mathrm{POD}(x_i)f(x_i|\theta)f_\epsilon(z_i/x_i)/x_i}{P_f(\theta)} \times \prod_{i=n+1}^{m}\frac{(1-\mathrm{POD}(x_i))f(x_i|\theta)}{1-P_f(\theta)}.
\end{aligned} \tag{3}$$

Therefore the likelihood of the truly observed data (in-service inspections) is obtained by:

$$\ell(\mathbf{z}|\theta) \quad = \quad \prod_{i=1}^{n}P_f^{-1}(\theta)\int \mathrm{POD}(x_i)f(x_i|\theta)f_\epsilon(z_i/x_i)/x_i\mathrm{d}x_i. \tag{4}$$

3. MAXIMUM LIKELIHOOD MAXIMIZATION

Preparatory works showed that the likelihood of a mixture of Weibull noisy data and statistically perfect data (coming from destructive testing) can suffer from identifiability issues when the progressive censoring characterizing the observational process is not accounted for, as testified on Figure 1. It was found that when the all data suffer from noise measurement (typically $\rho = 20\%$), those identifiability issues become worse. Nevertheless, accounting for both effects, the likelihood (4) can be maximized. Since no explicit conditional solution can emerge from likelihood equations, numerical optimization routines must be used. Tests demonstrated that a Gaussian quadrature method [**?**] can yield good approximations of the integrals involved in the above likelihood, and can be successfully coupled with a standard optimization method as Nelder-Mead [2] which furthermore avoids to specify the Jacobian vector $\partial\ell(\mathbf{z}|\theta)/\partial\theta$, as required by typical descent gradient methods (e.g., Newton-Raphson algorithms).

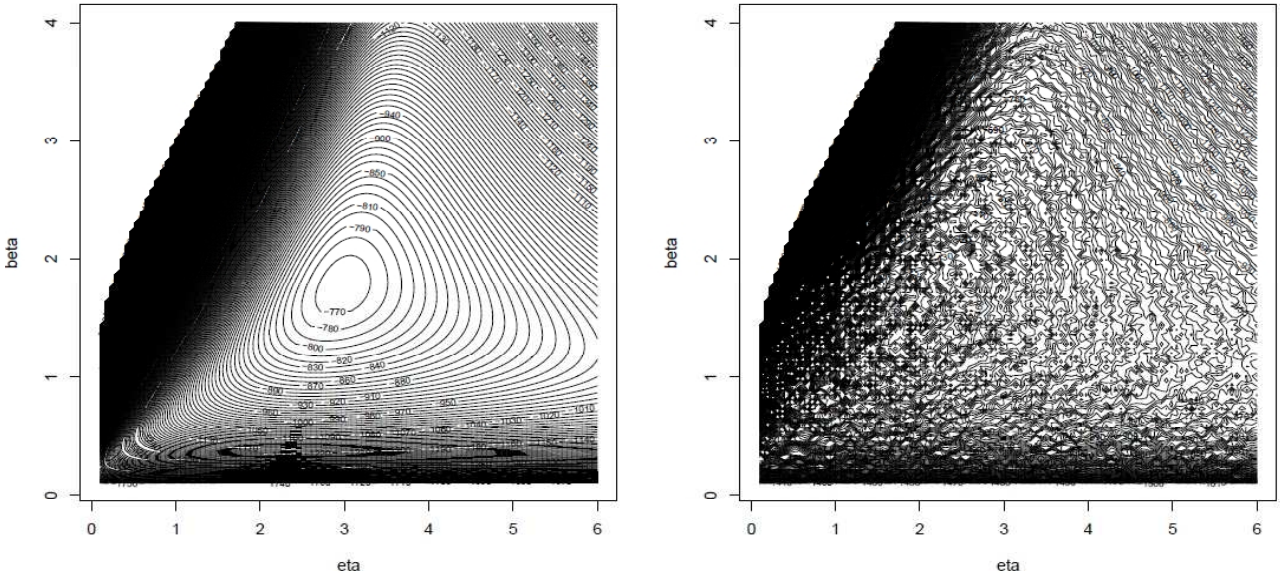


Figure 1: Level lines of the Weibull loglikelihood of $n = 430$ simulated data ($\beta = 1.8$, $\eta = 3.09$). *Left*: data are sampled without perturbation by observational noise, leading to an identifiable maxima. *Right*: 64% of the sampled data are affected by noise measurement ($\rho = 20\%$), which implies a strong loss of regularity implying identifiability issues of numerical optimization routines. The ratio of affected data is similar to those affecting truly the industrial data considered in this paper.

Nonetheless, such methods were found to be time-consuming and forbade the use of comfortably bootstrap methods for estimating the statistical features of the parameter estimators and the functions of interest, especially the probability $p(h)$ for a flaw height, observed through the noisy experimental process, to exceed the value h. In the experiments described at the following section, such bootstrap methods took nearly 66 hours on an AMD Sempron 3600. Denoting $\hat{p}(h)$ the MLE of $p(h)$, an alternative Delta method can be carried out [3], implying:

$$\mathrm{Var}[\hat{p}(h)] \quad \approx \quad \nabla F(h|\widehat{\theta})^{\top}\widehat{\Sigma}\nabla F(h|\widehat{\theta})$$

when $\widehat{\theta} = (\hat{\eta}, \hat{\beta})$ denotes the maximum likelihood estimator (MLE), $\widehat{\Sigma}$ is the inverse of the estimated Fisher information and F is the cumulative distribution function (cdf)

$$F(h|\theta) \quad = \quad \frac{\int \mathrm{POD}(x) f(x|\theta) F_{\epsilon}(h/x) dx}{P_f(\theta)} \tag{5}$$

with $F_{\epsilon}(h/x) = \int_0^h f_{\epsilon}(u/x)\, du$. The derivatives involved in the calculus of $\mathrm{Var}[\hat{p}(h)]$ (providing confidence intervals around $p(h)$) can easily be computed using numerical integration.

4. CALIBRATING THE PROBABILITY OF DETECTION (POD)

The POD is the most common indicator for quantifying the reliability and sensitivity of non-destructive in-service inspections [4, 5]. Its calibration appears as one of the most important methodological issues encountered in the area of performance testings, especially in nuclear contexts [6]. Such calibrations are based on a set of experimental performance measures, that are more and more often completed with means of numerical simulation reproducing the measurement process itself [7], especially to improve the validation and certification of these processes in industrial contexts or/and the training of manipulators [8, 9]. In this paper, we focalize on the performance measures. They can be either destructive, requiring costly ground samples, or non-destructive when benefiting from mock-ups that are machined with artificial flaws. In standard testing conditions, the certification of the ultrasonic process allowed to establish the precise limits of non-detection (POD=0%) and full detection (POD=100%), and to carry out a *signal response* method [10, 11] ; the logarithms of X and the signal S are linearly linked with normally-distributed random residuals ϵ':

$$\log S \quad = \quad \beta_0 + \beta_1 \log X + \epsilon',$$

which leads to the parametric form promoted by Berens [11]:

$$\mathrm{POD}(x) \quad = \quad \Phi\left(\frac{\log(x) - \log(x^*_m)}{1/q}\right) \tag{6}$$

where x^*_m is the height for which 50% of defects are detected, and q measures the quality of the POD (directing the average slope of the detection curve). However, the non-evolutive feature of the defects considered here does not allow to establish precise POD curves that are representative of the ground behavior as it could be done, for instance, when dealing with corrosion effects [12] inducing that successive measures can yield information. Therefore expert knowledge was introduced into the modelling to account for undesirable effects (e.g., due to false alarms) and form constraints (convexity properties of the POD).

5. SIMULATION EXPERIMENTS

In the experiments summarized in this section, the simulated data were produced as follows:

- $m_E = 190$ data assimilated to *destructive measures* follow the Weibull distribution $\mathcal{W}(\eta_0 = 3.09, \beta_0 = 1.8)$;
- $m_{IS} = 340$ data assimilated to *non-destructive measures* are produced using the same distribution but are submitted to a observational multiplicative noise with coefficient of variation $\rho = 20\%$ and the probability of detection (6) calibrated with $x_m = 6$ and $q = 20$ (this POD is plotted on Figure 4). This simulation can be easily produced by the following accept-reject algorithm: for $i = 1, \ldots, m_{IS}$,
  1. `sample` $x_i \sim f_W(.|\eta_0, \beta_0)$ `and` $u_i \sim \mathcal{U}_{unif}[0,1]$;
  2. `produce` $z_i = x_i \cdot \epsilon_i$ `with` $\epsilon_i \sim f_\epsilon$;
  3. `accept` z_i `if` $u_i \leq \int_{\mathbb{R}^+} \mathtt{POD}(z_i/u) f_\epsilon(u)\ du$.

It must be noticed that the phenomenon of detection produces non-destructive size measures that are significantly higher than the destructive measures, so that the former usually provide a truncated image of the common distribution through its upper tail. An example of sampled data and estimated distributions IS plotted on Figure 2. The MLE of (η, β) and $P_f(\theta)$ based on the mixture of destructive and non-destructive measures appeared to be a good estimator of these quantities, as well as the confidence intervals which were produced, as illustrated on Figure 3.

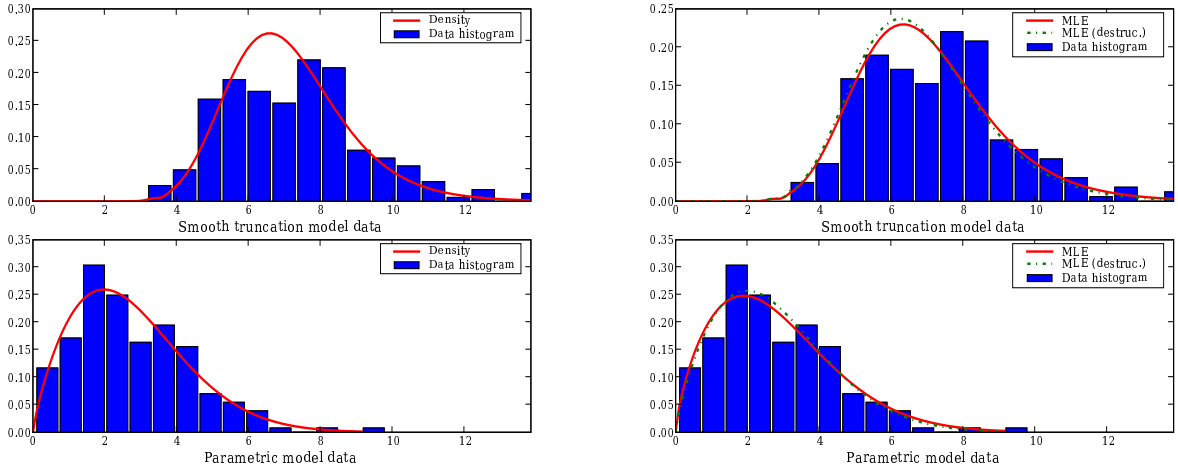


Figure 2: Histograms of simulated data. *Top*: non-destructive test data. *Down*: destructive test data. *Left*: true density function (solid line). *Right*: density function estimated by maximum likelihood on the basis of the complete data (MLE) or the destructive test data only (MLE (destruc.))

The variations between the MLE results based on the mixture of destructive and non-destructive data and the MLE based on the destructive data alone were always found to be statistically negligible. This feature allows to foresee the possibility of checking and strengthening the weakiest aspects of the modelling, namely the parametric model chosen for the probability of detection.

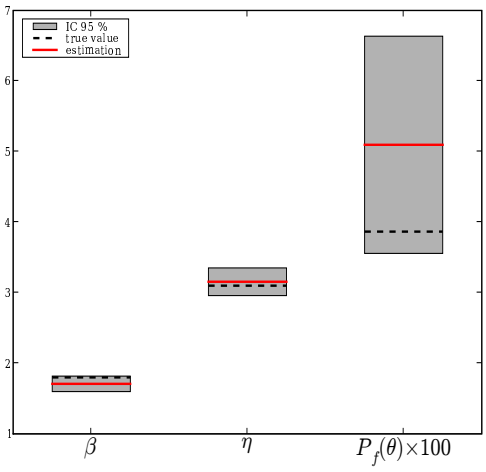


Figure 3: Estimates and 95% confidence intervals on (η, β) and $P_f(\theta)$, based on a mixture of non-destructive and destructive test data.

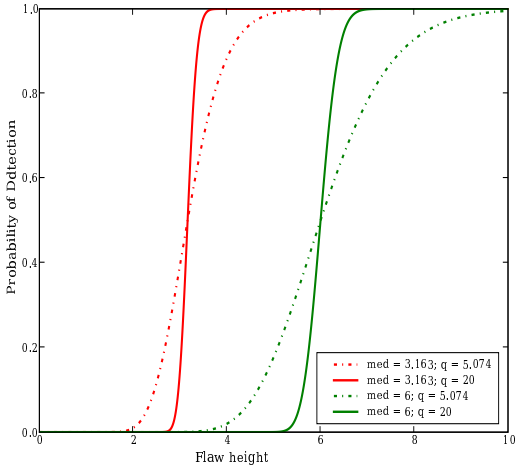


Figure 4: Two families of probability of detection. "med" is for the median fractile x_m^* in Equation (6).

6. APPLICATION TO PWR COMPONENT FLAWS: CHECKING THE RELEVANCE OF THE POD

The real available data are componed with 190 flaw observed on fractions of a steel component submitted to destructive laboratory experiments, and 341 couples of defect size (height and length) produced by ground experiments involving a ultrasonic device. Prior knowledge about the feature of this device was available to establish a series of four PODs (Figure 4). The differences between the PODs resulted from considerations about extreme behaviors (idealistic and pessimistic ground conditions)

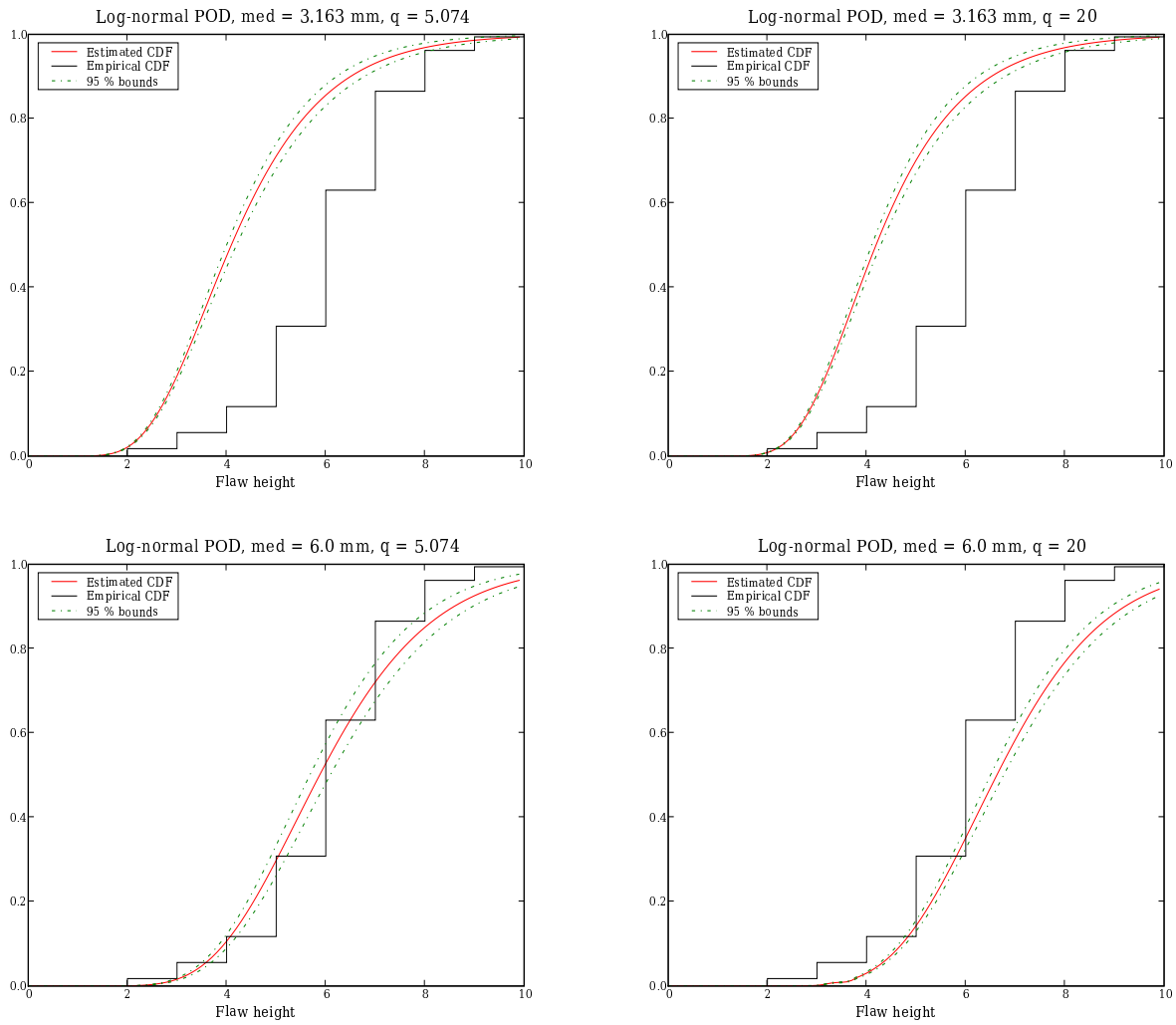


Figure 5: Estimation of the cumulative distribution function (cdf) of noisily observed flaw sizes as a function of the choice of POD, compared to their empirical cdf.

and variations of the state of the component surface. Supplementary information coming from the certification of the device was not used to establish these PODs, but was kept to offer a critical view of the estimation results.

The estimated cumulative distribution function of noisily observed flaw sizes, for each POD, are plotted on Figure 5. It can be seen that the POD calibrated with $x_m^* = 6$ and $q = 5.074$ provides the better fit with the empirical distribution function of the noisily observed flaw size, after a likelihood maximization involving the two databases. This POD indeed is the most in accordance with the technical knowledge arising from experiments on mock-ups. Especially, in the converted scale used in this article, the theoretical complete detection limit (in standard experimental conditions) was certified at the value $x = 8.5$.

7. CONCLUSIONS AND PROSPECTIVE AVENUES

À faire

- on peut estimer avec ou sans mlange

- fragilit = POD
- influence potentielle importante dans un cadre d'incertitude ?
- tape suivante = l'intgrer au maximum de vraisemblance
- l'estimer en tenant compte de toutes les connaissances

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